All nine four-variable quadratic monomial Hilbert series

Principia Math In Progress

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MathDB 360643. Complete intended all-power result. All nine conjectured Hilbert series are proved over every field. We also compute every Hilbert coefficient with the source's sign convention. Two resulting negative third coefficients disprove the broader nonnegativity conjecture, MathDB 360640. This consequence belongs to the same theorem and receives no separate campaign credit.

Task 1: the theorem and complete coefficient table

Let $S = k[x_1, x_2, x_3, x_4]$, let G be a graph on $V = \{1, 2, 3, 4\}$, and set

$$I_G = (x_1^2, x_2^2, x_3^2, x_4^2,\ x_i x_j : \{i, j\} \in E(G)).$$

In tables, ij denotes the edge $\{i, j\}$, and $s \geq 1$ is an integer. We use $\binom{a}{b} = 0$ when $0 \leq a < b$.

Theorem 1. *For each of the following nine graphs and every $s \geq 1$,*

$$\operatorname{Hilb}(S/I_G^s; t) = \sum_{d=0}^{2s-1} \binom{d+3}{3} t^d + A_G(s) t^{2s} + B_G(s) t^{2s+1}, \tag{1}$$

where A_G, B_G are given in Table 1. The complete Hilbert coefficient vectors are those in Table 2, with convention

$$\ell(S/I_G^s) = \sum_{j=0}^{4} (-1)^j e_j(I_G) \binom{s+3-j}{4-j}. \tag{2}$$

These formulas hold for all positive powers, not only eventually.

Row	$E(G)$	$A_G(s)$	$B_G(s)$
1	12	$s(s+1)(2s+3)/2$	$2\binom{s+2}{3}$
2	12, 13	$4\binom{s+2}{3}$	$\binom{s+1}{2}$
3	12, 34	$4\binom{s+2}{3}$	0
4	12, 13, 23	$s(s+1)(4s+5)/6$	0
5	12, 13, 14	$s(2s+1)$	$\binom{s+1}{2}$
6	12, 23, 34	$s(2s+1)$	0
7	12, 23, 34, 14	$2s$	0
8	12, 13, 14, 23	$s(s+1)$	0
9	12, 13, 14, 23, 24	s	0

Table 1. All nine source Hilbert-series formulas.

Row	e_0	e_1	e_2	e_3	e_4	Row	e_0	e_1	e_2	e_3	e_4
1	16	4	0	0	0	6	16	12	5	1	0
2	16	8	2	0	0	7	16	12	1	−2	0
3	16	8	1	0	0	8	16	12	3	0	0
4	16	8	0	0	0	9	16	12	1	−1	0
5	16	12	6	1	0						

Table 2. Complete coefficient vectors in convention (2).

Compared with the source, e_1 in rows 2 and 3 is 8, the omitted e_3's in rows 5 and 6 are 1, and the signs of e_3 in rows 7 and 9 are negative. The main result is the proof of every series for every power. These arithmetic corrections alone earn no credit.

Monomial membership and the degree bound

Write $a_i = 2b_i + \epsilon_i$, where $b_i \geq 0$ and $\epsilon_i \in \{0, 1\}$, for a monomial x^a. Let $O = \{i : \epsilon_i = 1\}$, and let $U = \{i : a_i > 0\}$ be its support. A product of quadratic generators can be regarded as pairs of tokens: equal-index pairs use squares and distinct-index pairs must be edges of G. Thus $x^a \in I_G^s$ exactly when at least s such pairs can be chosen within its exponent capacities.

All monomials of degree below $2s$ survive, giving the first sum of (1). Every monomial of degree $2s+2$ lies in the ideal. Indeed $|O|$ is 0, 2 or 4. If it is 0 or 2, the squares alone give at least s pairs. If it is 4, use any edge of G (all nine graphs are nonempty); after removing its two tokens, the remaining squares give $s-1$ more pairs. Every higher-degree monomial is a multiple of a degree-$2s+2$ monomial, so it also lies in I_G^s.

Counting degree $2s$

When $|O| = 0$, squares give an exact pairing, so there are no survivors. Suppose $O = \{u, v\}$. We claim that

$$x^a \notin I_G^s \quad \Longleftrightarrow \quad u, v \text{ lie in different components of } G[U]. \tag{3}$$

For necessity of connectivity, an exact pairing uses only edges inside support components; each such component must contain an even total number of tokens. If u, v are separated, their two components have odd totals, making a pairing impossible. Conversely, if they are connected, choose a simple path from u to v in $G[U]$, using each path edge once. Every internal vertex has a positive even exponent, hence at least two available tokens. The endpoints have at least one. The remaining exponents are all even and can be paired using squares. This proves (3).

Fix O and a set $T \subseteq V \setminus O$ of the positive even-exponent vertices. Then $U = O \cup T$, $\sum b_i = s - 1$, $b_i \geq 1$ on T, and $b_i = 0$ outside $O \cup T$. Stars and bars gives $\binom{s}{|T|+1}$ choices; it gives zero automatically if there are too few tokens. Let $c_t(G)$ count those pairs (O, T) with $|O| = 2$, $|T| = t$ and separated odd vertices. Thus the two-odd-exponent contribution is

$$c_0\binom{s}{1} + c_1\binom{s}{2} + c_2\binom{s}{3}. \tag{4}$$

These small graph counts can be obtained without ideal computations. For any induced graph H on m vertices, its number of separated pairs is $\binom{m}{2} - \sum_C \binom{|C|}{2}$, summing over its components. Consequently $c_0 = 6 - |E(G)|$, c_1 is the sum of this count over the four induced triples, and c_2 is the count on G itself.

It remains to count $|O| = 4$, when $\sum b_i = s - 2$. Rows 1, 2 and 4 have an odd-size support component, so no exact pairing is possible. All $\binom{s+1}{3}$ monomials survive. Rows 3, 6, 7, 8 and 9 have a perfect matching; use that matching and then squares, so no monomial survives. In row 5 the graph is a star centered at 1. The three odd leaves each require an odd, hence positive, number of incident edges in an exact pairing. Thus at least three center tokens are necessary. They suffice: use each star edge once, and pair the rest by squares. The survivors therefore have $a_1 = 1$, equivalently $b_1 = 0$, and there are $\binom{s}{2}$ of them. Let $F_G(s)$ be this four-odd contribution. We have proved

$$A_G(s) = c_0\binom{s}{1} + c_1\binom{s}{2} + c_2\binom{s}{3} + F_G(s). \tag{5}$$

Row	c_0	c_1	c_2	$F_G(s)$	d_0	d_1
1	5	10	5	$\binom{s+1}{3}$	2	0
2	4	7	3	$\binom{s+1}{3}$	0	1
3	4	8	4	0	0	0
4	3	6	3	$\binom{s+1}{3}$	0	0
5	3	3	0	$\binom{s}{2}$	0	1
6	3	4	0	0	0	0
7	2	0	0	0	0	0
8	2	2	0	0	0	0
9	1	0	0	0	0	0

Table 3. Counts used in (5) and (6).

Counting degree $2s + 1$

There are either one or three odd exponents. One odd exponent leaves s square pairs, so gives no survivor. With three odd exponents, squares initially give only $s - 1$ pairs. An edge between two odd vertices supplies the missing pair. If the odd set O is independent, only the fourth vertex v can connect two odd vertices. If v has at least two neighbors in O and $a_v \geq 2$, the length-two path through v does supply s pairs, leaving the third odd token unpaired.

In the other cases the three odd vertices lie in three distinct support components. A pairing leaving only one token would have to leave a token in each of these three odd components, which is impossible. This proves the necessity as well as the sufficiency of the preceding criterion.

Let d_0 count independent triples whose remaining vertex has at most one neighbor in the triple, and d_1 those whose remaining vertex has at least two neighbors. In the first case all four b_i's are free with sum $s - 1$, giving $\binom{s+2}{3}$ survivors. In the second, survival forces $b_v = 0$, giving $\binom{s+1}{2}$. Hence

$$B_G(s) = d_0\binom{s+2}{3} + d_1\binom{s+1}{2}. \tag{6}$$

Row 1 has two such independent triples of the first type. Rows 2 and 5 each have the single triple $\{2, 3, 4\}$ of the second type. The other rows have no independent triple. This gives the last two columns of Table 3. Substitution into (5)–(6) gives every entry of Table 1, completing the proof of (1).

Hilbert coefficients and the positivity counterexample

Summing (1) gives the exact polynomial

$$\ell(S/I_G^s) = \binom{2s+3}{4} + A_G(s) + B_G(s). \tag{7}$$

The five binomial polynomials in (2) have distinct degrees, so their coefficients are unique. Expanding (7) in that basis gives Table 2. For example, direct polynomial expansion gives

$$\binom{2s+3}{4} = 16\binom{s+3}{4} - 12\binom{s+2}{3} + \binom{s+1}{2}.$$

Thus row 7, the four-cycle, has

$$\ell(S/I_G^s) = 16\binom{s+3}{4} - 12\binom{s+2}{3} + \binom{s+1}{2} + 2s,$$

so $e_3(I_G) = -2$ in the source's convention. This ideal is Artinian because it contains every square, and is generated entirely in degree two. It therefore disproves Conjecture 1 in the preprint, Conjecture 1.1 in the final (MathDB 360640), with all its hypotheses satisfied. Row 9 likewise gives $e_3 = -1$. These are consequences of the proved full series, not inferences from the source's unproved formulas or from a sign typo alone.

Tasks 2–5: four proposed follow-ups

Task 2: Extend the parity and support-graph classification to all five-variable ideals generated by squares and graph edges.

Task 3: Characterize the graphs for which all Hilbert coefficients of the corresponding quadratic ideal are nonnegative.

Task 4: Determine the depths of the associated graded rings for the nine ideals in Table 1.

Task 5: Compare their ordinary-power coefficients with those of the integral-closure filtrations, explaining when the negative signs disappear.

These are proposed directions, not additional claimed solutions.

Sources and verification

R. Fröberg, *Hilbert coefficients of quadratic algebras*, arXiv:2311.02860v1, 6 November 2023, Conjecture 4, p. 6; all seven pages read. Final: *Journal of Experimental Mathematics* 1(2) (2025), 315–321, DOI 10.56994/JXM.001.002.007, Conjecture 3.2, pp. 319–320; all seven article pages and both front-matter pages read.

The source supplies the nine conjectured series. Elementary monomial membership and stars and bars are the inputs to our all-power proof. Negative third coefficients for general ideals were already known; the counterexamples here satisfy the additional quadratic hypothesis.

Three supplementary exact groups check 89,856 monomials, 216 graded counts, the graph counts, and coefficient recovery from lengths. One combined result earns one campaign credit. Review by the authoring agent only, without independent peer review or a proof assistant; bounded source and claim checks are recorded in the audit.